



Tribhuvan University
Faculty of Humanities and Social Science
Swastik college, Bhaktapur



Cloud Computing
Lab Report

Submitted By:

Swosti Makaju(113102097)

BCA 7th Semester

Submitted To:

Sujan Poudel

Lecturer, BCA Department

Swastik college, Bhaktapur

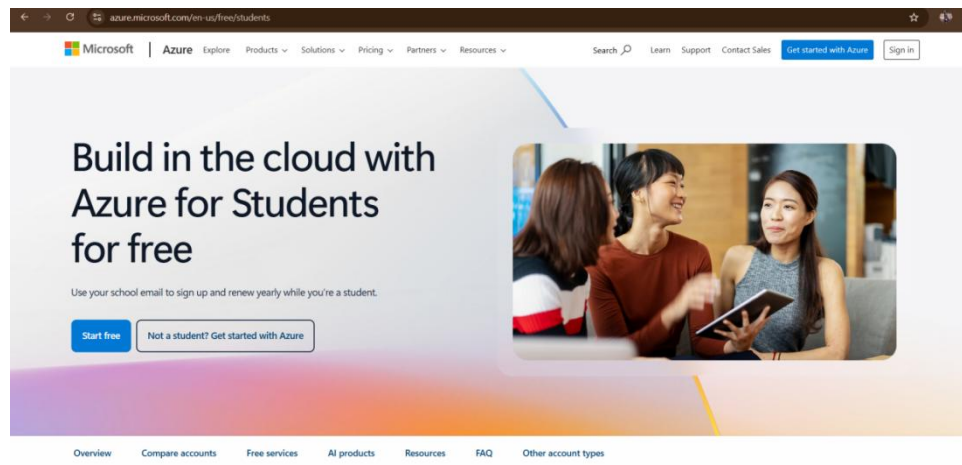
Lab 1: Getting Started with Azure

Objective: Understand the Azure portal, subscription model, and cloud terminology.

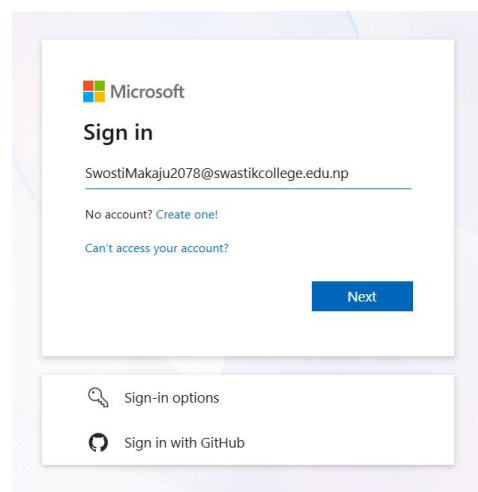
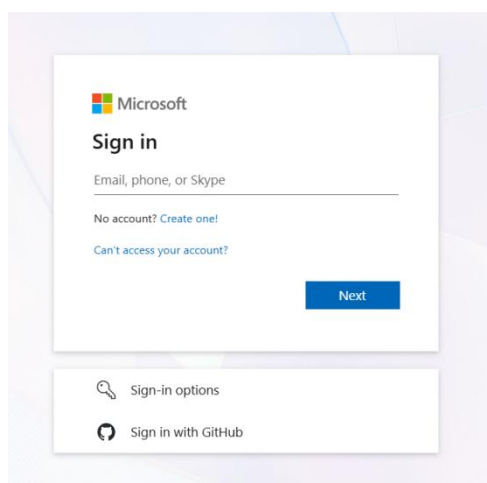
Tasks:

1. Create an Azure Account

- Visit: <https://azure.microsoft.com/en-us/free/students>



- Click **Start Free** and sign in with a Microsoft account (use a school/university email).

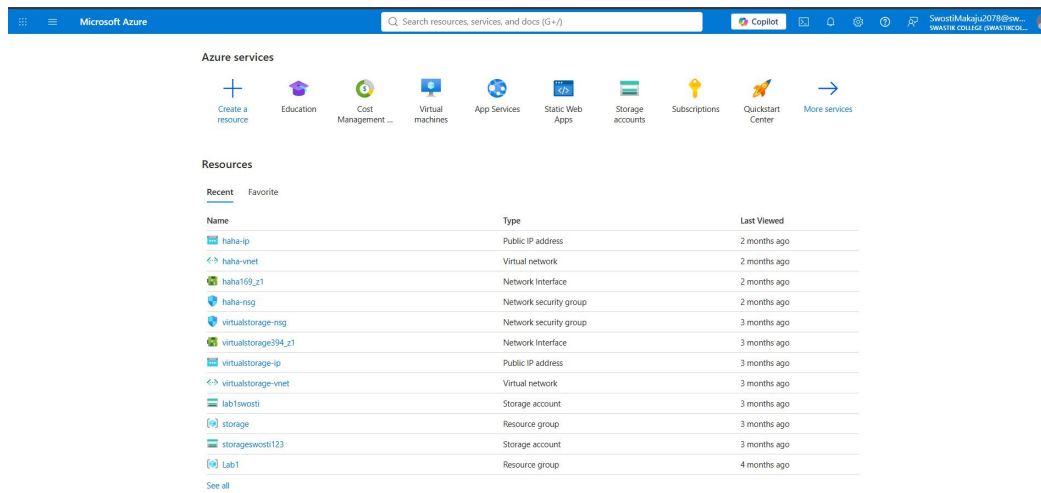


- Select **Student Option** to get \$100 credits via [Azure for Students](#).
- Complete the sign up and verification (phone number + identity verification).

Note: No credit card is needed for student accounts.

2. Explore Azure Portal and Cloud Shell

- Login to the [Azure Portal](#).



- Familiarize with **Dashboard, Resource Groups, All Services, Marketplace**

3. Create a Resource Group

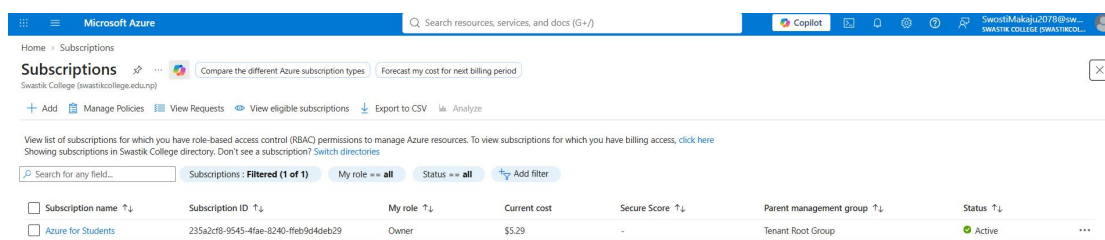
A resource group is a logical container for your Azure resources.

- **Via Portal:**

- Go to **Resource Groups** → **Create**
- Resource Group Name: **lab1swosti**
- Region: **South India** (or nearest region)

4. Verify Azure Subscription

- Go to Subscriptions and verified the Azure for students subscription.
- Confirmed that the subscription status was Active.



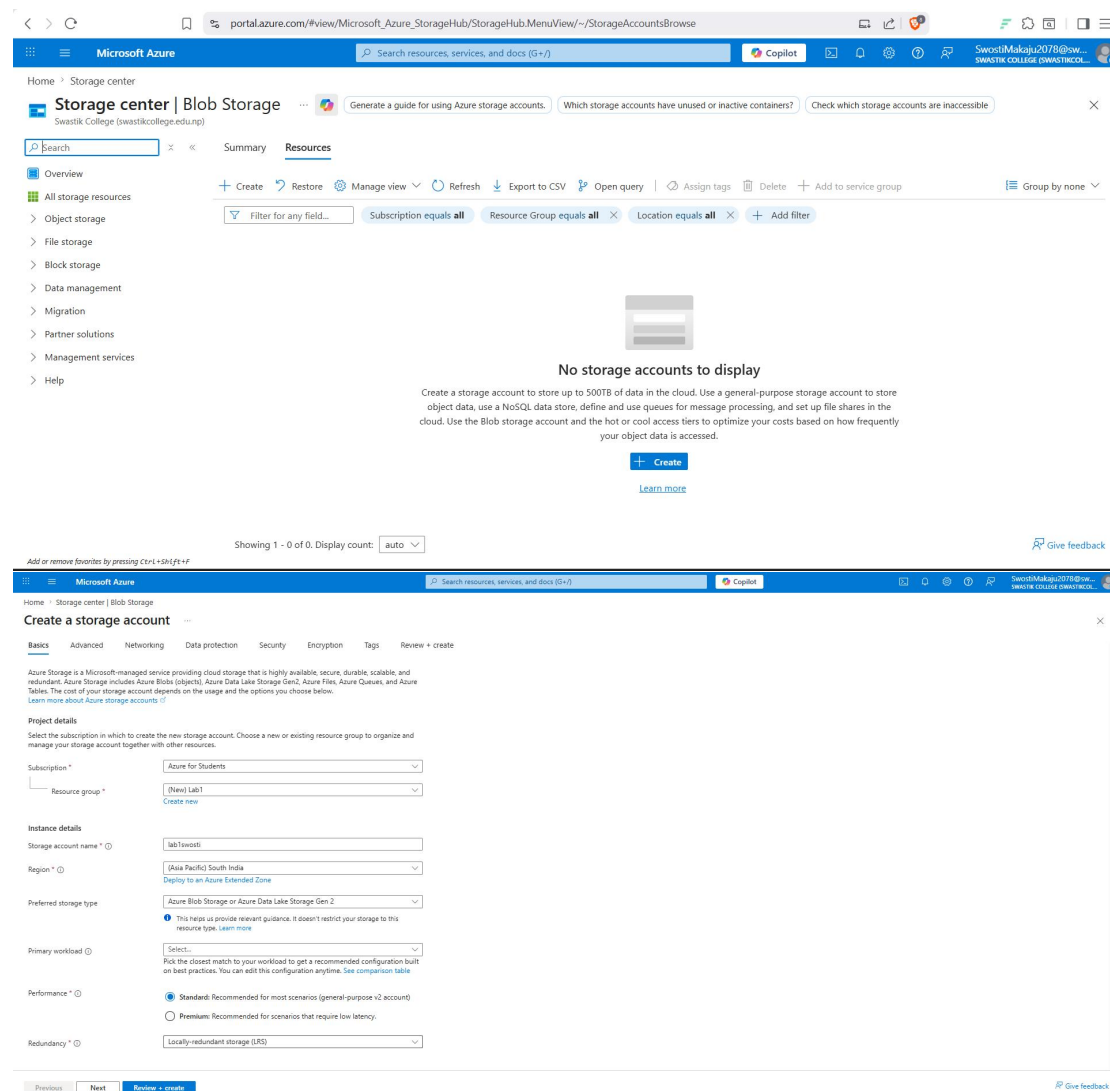
Lab 2: Azure Storage Services

Objective:

Learn to create and manage Azure Blob Storage, including uploading files and accessing them via public URLs using both the Portal and Azure CLI.

Via Portal:

1. Go to Storage accounts → Create



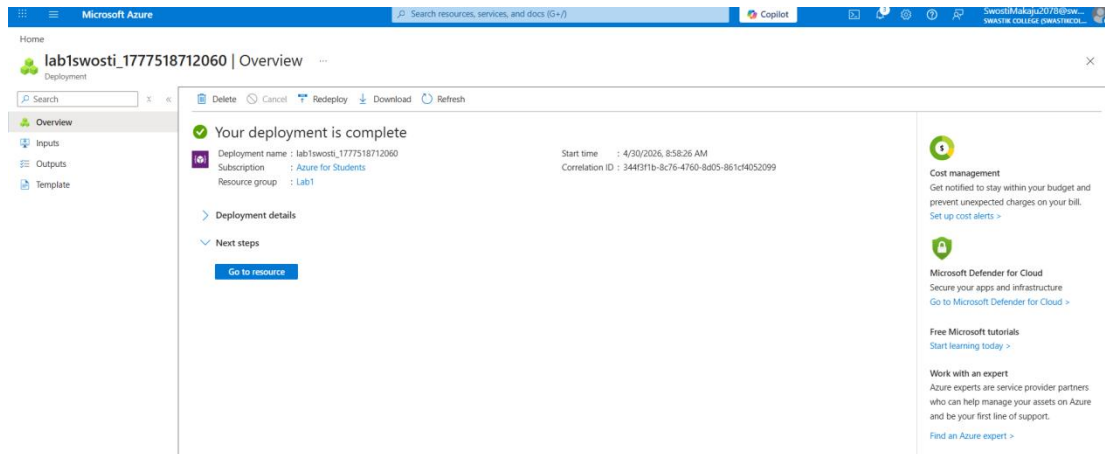
The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is titled 'Storage center | Blob Storage'. It features a left-hand navigation pane with options like Overview, All storage resources, Object storage, File storage, Block storage, Data management, Migration, Partner solutions, Management services, and Help. The main area displays a message: 'No storage accounts to display'. Below this message, there is a '+ Create' button and a 'Learn more' link. The page also includes a description of Azure Storage and a 'Showing 1 - 0 of 0. Display count: auto' message.

2. Fill in the details:

- Subscription: (Select your active subscription)
- Resource Group: **Lab1** (or create a new one)
- Storage Account Name: **lab1swosti**

- Region: South India
- Performance: Standard
- Redundancy: Locally-redundant storage (LRS)

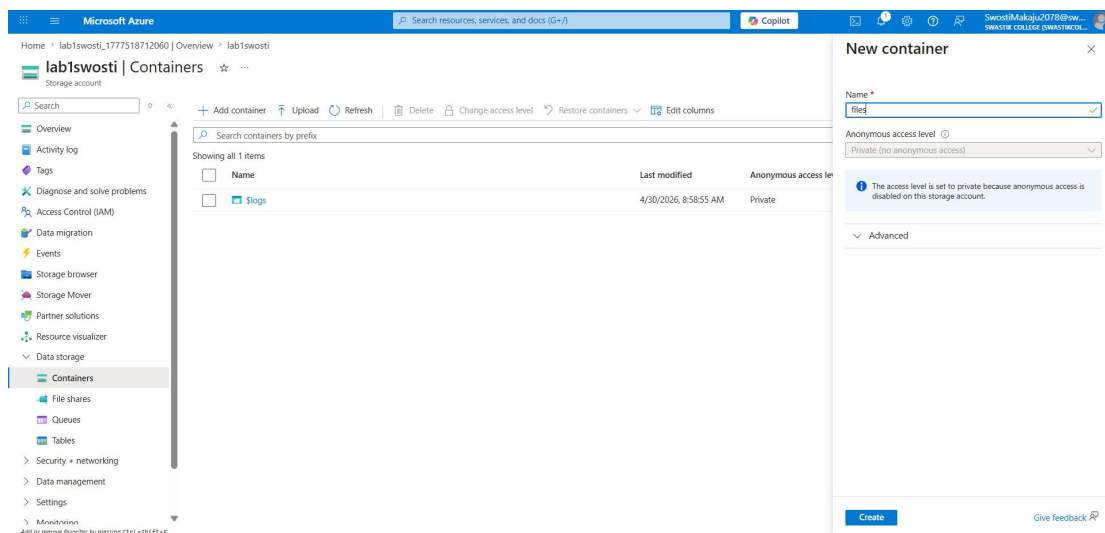
3. Click **Review + Create**, then **Create**



4. Create a Blob Container

Via Portal:

- Navigate to your newly created Storage Account
- Under **Data storage**, select **Containers**
- Click **+ Container**



Lab 3: Azure Virtual Machines

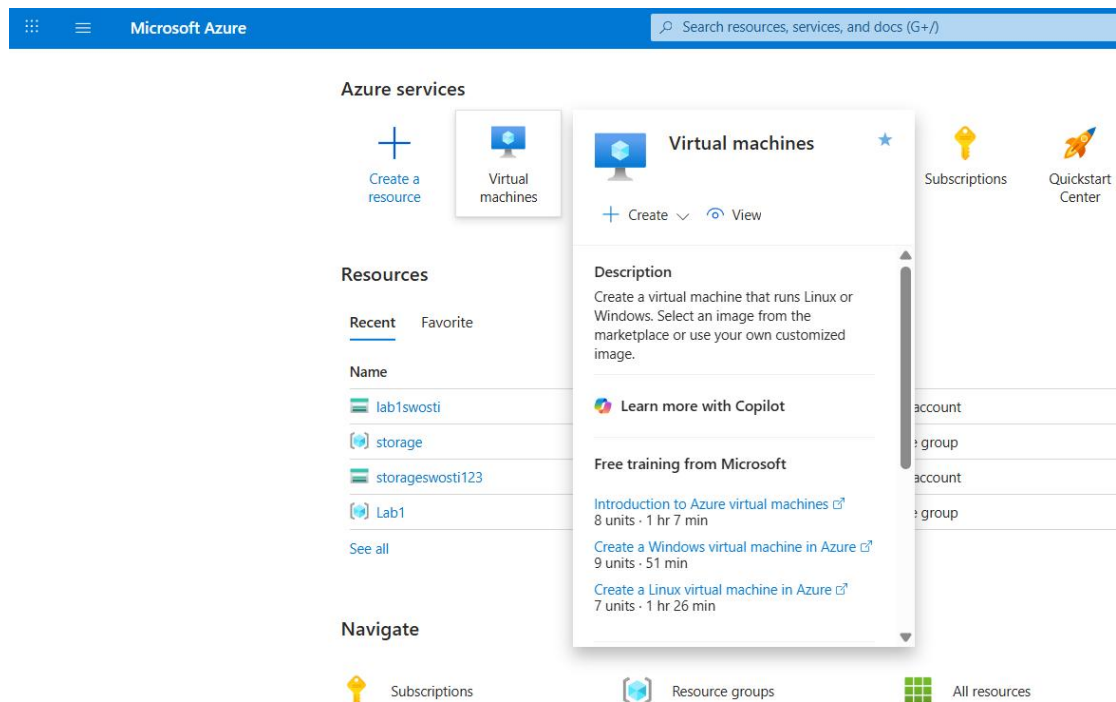
Objective: Deploy a virtual machine and perform basic server tasks.

Deploy a Linux virtual machine on Azure, connect to it via SSH, and host a static website using Apache or Nginx.

1. Create a Linux-based VM

Via Portal:

1. Go to Virtual Machines → Create



2. Fill in the following:

- Subscription: Your active one
- Resource Group: storage
- VM name: **virtualstorage**
- Region: East US (or nearby)

Microsoft Azure

Q

Search resources, services, and docs (G+/)

Home

>

Compute infrastructure | Virtual machines

Create a virtual machine

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Subscription * ⓘ

Azure for Students

▼

Resource group * ⓘ

storage

▼

Create new

Instance details

Virtual machine name * ⓘ

virtualstorage

Region * ⓘ

(Asia Pacific) East Asia

▼

Deploy to an Azure Extended Zone

Availability options ⓘ

Availability zone

▼

Zone options * ⓘ

☒ Self-selected zone
 Select a specific zone for your VM

☐ Azure-selected zone (Preview)
 Let Azure assign the best zone for your needs

Availability zone * ⓘ

Zone 1

▼

Security type ⓘ

Trusted launch virtual machines

▼

Configure security features

Image * ⓘ

Ubuntu Server 24.04 LTS - x64 Gen2

▼

See all images | Configure VM generation

Previous

Next

Review + create

- Image: Ubuntu Server 20.04 LTS
- Size: Standard B1s (Free tier if available)

Security type ⓘ

Trusted launch virtual machines

▼

Configure security features

Image * ⓘ

Ubuntu Server 24.04 LTS - x64 Gen2

▼

See all images | Configure VM generation

VM architecture ⓘ

☐ Arm64
 ☒ x64

Run with Azure Spot discount ⓘ

☒

Eviction type ⓘ

☒ Capacity only: Your virtual machine will be evicted when Azure's excess capacity disappears.
 ☐ Price or capacity: Your virtual machine will be evicted when Azure's excess capacity disappears, or costs exceed your specified max price.

Eviction policy ⓘ

☒ Stop / Deallocate
 ☐ Delete

Size * ⓘ

Standard_D2s_v3 - 2 vcpus, 8 GiB memory (\$0.01940)

▼

See all sizes | Customize cores

View pricing history and compare prices in nearby regions

- Authentication: SSH (generate or upload key)
- Allow inbound ports: SSH (22), HTTP (80)

Administrator account

Authentication type * ⓘ

☐ SSH public key

☒ Password

Username * ⓘ

swosti

Password * ⓘ

Confirm password * ⓘ

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☐ None

☒ Allow selected ports

Select inbound ports *

HTTP (80), SSH (22) ▾

ⓘ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

3. Click Review + Create, then Create

Microsoft Azure

Search resources, services, and docs (G+)

Home > Compute infrastructure | Virtual machines

Create a virtual machine

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

[View automation template](#)

Price

1 X Standard D2s_v3
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
0.0244 USD/hr

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

SwostiMakaju2078 SwostiMakaju2078

Preferred e-mail address

SwostiMakaju2078@swastikcollege.edu.np

Preferred phone number

16635174

4. After deployment, go to the VM → Note down the Public IP address then host a static website

virtualstorage

Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Connect ▾ ▶ Start Restart Stop ⓘ Hibernate Capture ▾ Delete Refresh ⚙ Scale Open in mobile Feedback 🗨️ CU / PS

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Resource group: default

Essentials

Resource group (move) : storage

Status : Running

Location : East Asia (Zone 1)

Subscription (move) : Azure for Students

Subscription ID : 235a2cf8-9545-4fae-8240-f1eb9d4deb29

Availability zone : 1

Tags (edit) : Add tags

Properties

Virtual machine

Computer name : virtualstorage

Operating system : Linux (ubuntu 24.04)

VM generation : V2

VM architecture : x64

Monitoring

Capabilities (7)

Recommendations

Tutorials

Operating system : Linux (ubuntu 24.04)

Size : Standard D2s_v3 (2 vcpus, 8 GiB memory)

Primary NIC public IP : 104.214.184.0

1 associated public IPs

Virtual network/subnet : vnet-eastasia-1/vnet-eastasia-1

DNS name : Not configured

Health state : -

Time created : 8/23/2026, 6:28 AM UTC

Networking

Public IP address ⓘ : 104.214.184.0 (Network interface virtualstorage581)

1 associated public IPs

Public IP address (IPv6) : -

Private IP address : 172.16.0.4

5. Host a Static Website

```
swosti@virtualstorage: ~ × Settings × + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\swost>ssh swosti@104.214.184.0
swosti@104.214.184.0's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1022-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Aug 23 06:37:15 UTC 2026

System load:  0.08          Processes:            120
Usage of /:   5.6% of 28.02GB Users logged in:        0
Memory usage: 3%           IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

swosti@virtualstorage:~$ |
```

Lab 4 NGINX Configuration

1. Do SSH connection with your virtual machine

Here the virtual machine name is “newvm” and the ip is “4.240.110.169” . After doing ssh enter password.

```
ssh <swosti>@<4.240.110.169>
```

```
ssh swosti@4.240.110.169
```

```
Swosti12345678
```

```
C:\Users\swost>ssh swosti@4.240.110.169
The authenticity of host '4.240.110.169 (4.240.110.169)' can't be established.
ED25519 key fingerprint is SHA256:6xZhTJ+uqHLRnTXtFulltAlz/FotfttCFhkm+KcmLc4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '4.240.110.169' (ED25519) to the list of known hosts.
swosti@4.240.110.169's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1022-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Aug 25 05:35:42 UTC 2026

System load:  0.08               Processes:            121
Usage of /:   5.6% of 28.02GB    Users logged in:     0
Memory usage: 3%                IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

swosti@newvm:~$ |
```

1. Install NGINX

Update system packages and install NGINX:

```
sudo apt update
```

```
swosti@newvm:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
```

sudo apt install nginx -y

```
swosti@newvm:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 35 not upgraded.
Need to get 568 kB of archives.
After this operation, 1602 kB of additional disk space will be used.
```

Verify that NGINX is running:

sudo systemctl status nginx

```
swosti@newvm:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-08-25 05:38:34 UTC; 50s ago
     Docs: man:nginx(8)
   Process: 2251 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 2253 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2282 (nginx)
    Tasks: 3 (limit: 9433)
   Memory: 2.4M (peak: 5.3M)
      CPU: 22ms
   CGroup: /system.slice/nginx.service
           └─2282 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2285 "nginx: worker process"
               └─2286 "nginx: worker process"

Aug 25 05:38:34 newvm systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server.
Aug 25 05:38:34 newvm systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

2. Connect GitHub to the Virtual Machine

If this is the first time connecting GitHub:

1. Generate SSH key: *ssh-keygen -t ed25519 -C"your_email@example.com"*

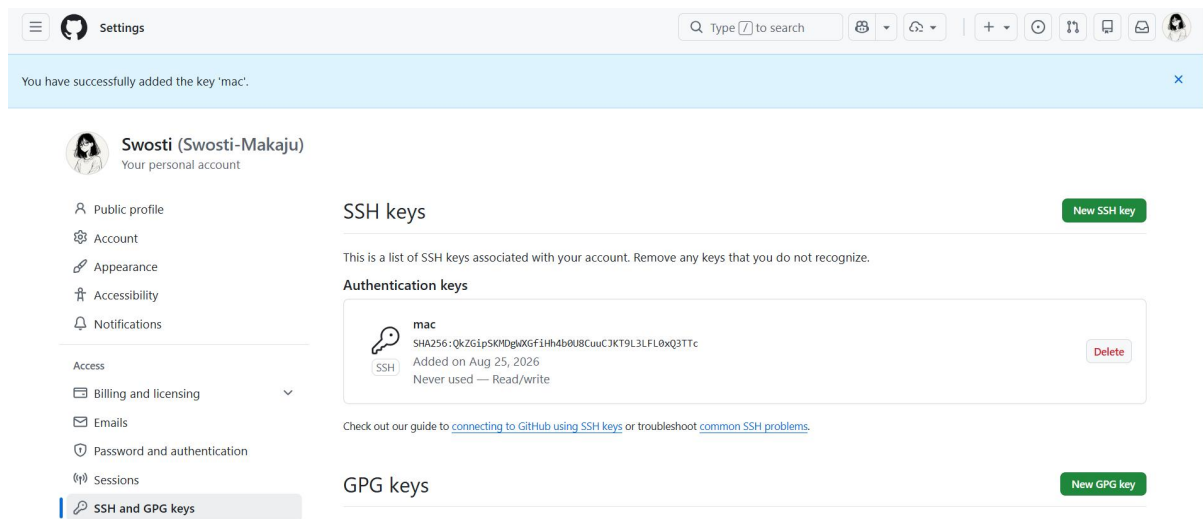
```
swosti@newvm:~$ ssh-keygen -t ed25519 -C "swosti.makaju@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/swosti/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/swosti/.ssh/id_ed25519
Your public key has been saved in /home/swosti/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:QkZGipSKMDgWXGfIHh4b0U8CuuCJKT9L3LFL0xQ3TTc swosti.makaju@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|o..+o+..      . E|
|==+.*. o . .|
|==.. B o .|
|*.. =o + .|
|++ +. o S|
|.o . = .|
|.. = .|
|. + o|
|. .|
+-----[SHA256]-----+
swosti@newvm:~$
```

2. View and copy your public key:

```
cat ~/.ssh/id_ed25519.pub
```

```
swosti@newvm:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIkI5Mqynxar30KKeNsYDHW6qHWLch1PsDhtRdhNq0l swosti.makaju@gmail.com
swosti@newvm:~$
```

3. Add it to GitHub → Settings → SSH and GPG Keys → New SSH Key.



Test the connection:

```
ssh -T git@github.com
```

```
swosti@newvm:~$ ssh -T git@github.com
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi Swosti-Makaju! You've successfully authenticated. but GitHub does not provide shell access.
```

3. Pull the Repository You Want to Host

Navigate to the directory where you want the project:

```
cd /var/www/
```

```
swosti@newvm:~$ cd /var/www
swosti@newvm:/var/www$
```

Clone the repository:

```
sudo git clone git@github.com:username/repository.git
```

If avoid permission error then do:

```
sudo chown -R swosti:swosti /var/www
```

```
swosti@newvm:/var/www$ sudo chown -R swosti:swosti /var/www
```

```
swosti@newvm:/var/www$ git clone https://github.com/Swosti-Makaju/Cloud_Computing.git
Cloning into 'Cloud_Computing'...
remote: Enumerating objects: 17, done.
remote: Counting objects: 100% (17/17), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 17 (delta 4), reused 15 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (17/17), 5.50 KiB | 5.50 MiB/s, done.
Resolving deltas: 100% (4/4), done.
```

4. Run the Project & Install Dependencies

Go into the project folder:

cd repository

```
swosti@newvm:/var/www$ cd Cloud_Computing
swosti@newvm:/var/www/Cloud_Computing$ ls
README.md  app.py  requirements.txt  templates
```

Install dependencies

Python (FastAPI / Django / Flask)

```
sudo apt update && sudo apt install -y python3-venv python3-full
pip install -r requirements.txt
```

```
swosti@newvm:/var/www/Cloud_Computing$ sudo apt update && sudo apt install -y python3-venv python3-full
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
33 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
swosti@newvm:/var/www/Cloud_Computing$ python3 -m venv venv
swosti@newvm:/var/www/Cloud_Computing$ source venv/bin/activate
(venv) swosti@newvm:/var/www/Cloud_Computing$ pip install -r requirements.txt
Collecting flask (from -r requirements.txt (line 1))
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting azure-storage-blob (from -r requirements.txt (line 2))
  Downloading azure_storage_blob-12.30.0-py3-none-any.whl.metadata (26 kB)
Collecting python-dotenv (from -r requirements.txt (line 3))
  Downloading python_dotenv-1.2.3-py3-none-any.whl.metadata (29 kB)
Collecting blinker>=1.9.0 (from flask->-r requirements.txt (line 1))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask->-r requirements.txt (line 1))
  Downloading click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask->-r requirements.txt (line 1))
```

6. Create an NGINX Site Configuration File

Go to NGINX's site-available directory:

cd /etc/nginx/sites-available/

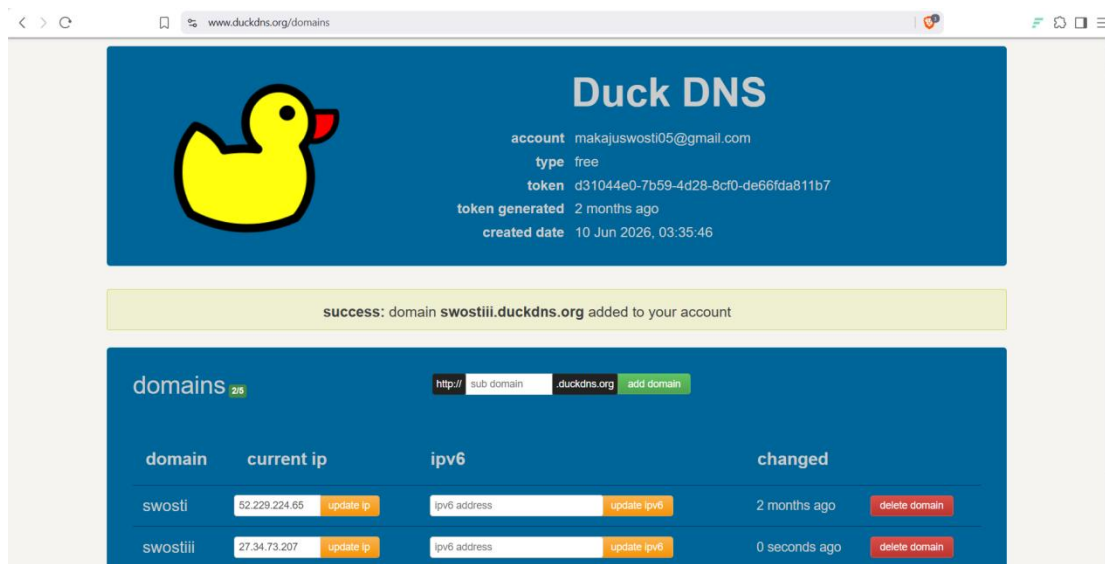
```
(venv) swosti@newvm:/var/www/Cloud_Computing$ cd /etc/nginx/sites-available/  
(venv) swosti@newvm:/etc/nginx/sites-available$ ls  
default
```

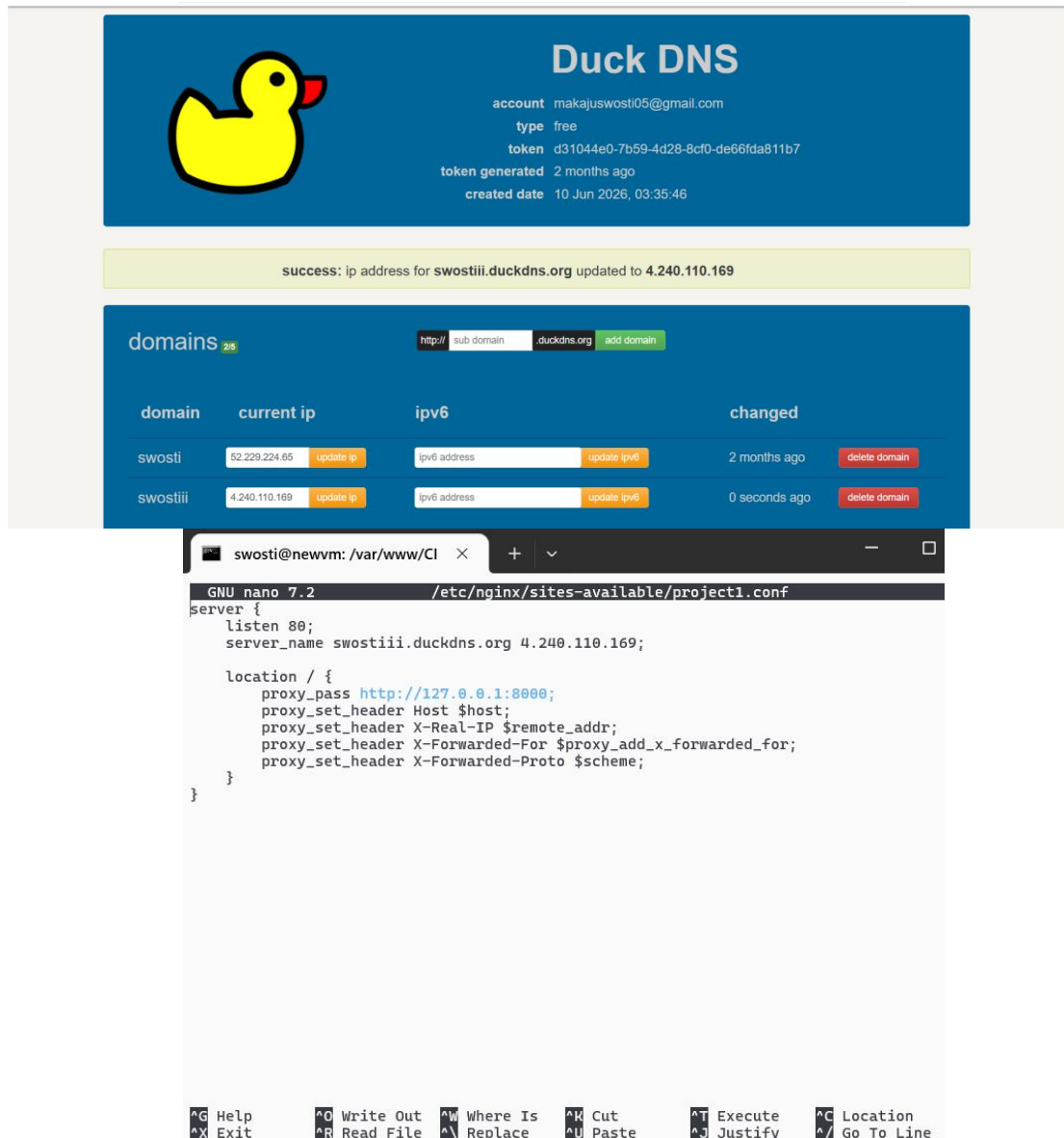
Create a config file for your project:

sudo nano project1.conf



We need a domain name for this purpose so we are using the free duckdns.org domain for the lab.





The image shows a screenshot of the Duck DNS website and a terminal window. The website displays account information for 'makajuswosti05@gmail.com' and a list of domains. The terminal window shows the configuration of an nginx server in the file `/etc/nginx/sites-available/project1.conf`.

Duck DNS Account Information:

- account: makajuswosti05@gmail.com
- type: free
- token: d31044e0-7b59-4d28-8cf0-de66fda811b7
- token generated: 2 months ago
- created date: 10 Jun 2026, 03:35:46

Success Message: ip address for swostiii.duckdns.org updated to 4.240.110.169

Domains Table:

domain	current ip	ipv6	changed
swosti	52.229.224.65	ipv6 address	2 months ago
swostiii	4.240.110.169	ipv6 address	0 seconds ago

Terminal Window (nano 7.2):

```
server {
    listen 80;
    server_name swostiii.duckdns.org 4.240.110.169;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

7. Create a Symlink to Enable the Site

Link your config from **sites-available** to **sites-enabled**:

sudo ln -s /etc/nginx/sites-available/project1.conf /etc/nginx/sites-enabled/

```
(venv) swosti@newvm:/etc/nginx/sites-available$ sudo ln -s /etc/nginx/sites-available/project1.conf /etc/nginx/sites-enabled/
(venv) swosti@newvm:/etc/nginx/sites-available$ |
```

8. Test NGINX Configuration

Check for syntax errors:

sudo nginx -t

```
(venv) swosti@newvm:/etc/nginx/sites-available$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

If everything is OK, proceed.

9. Reload or Restart NGINX

Apply the changes:

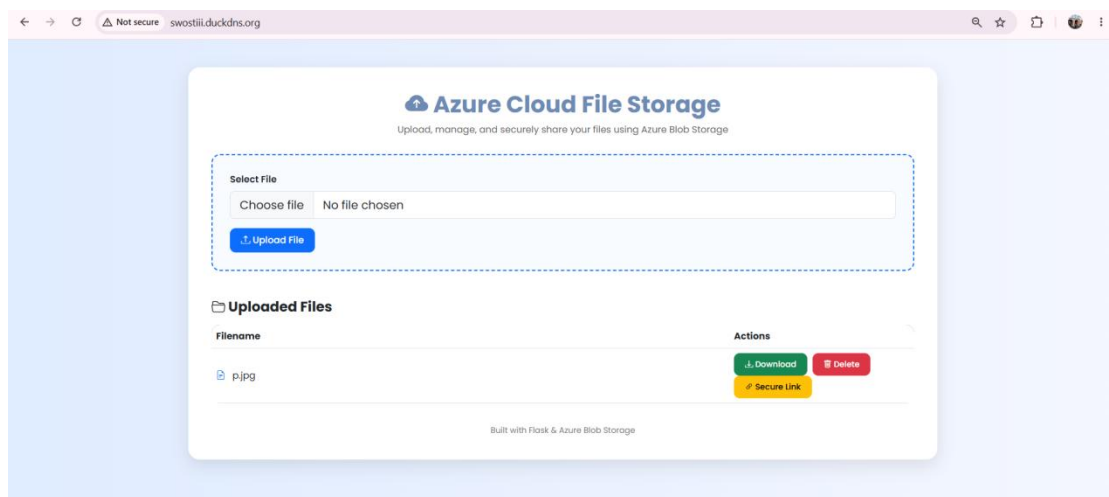
sudo systemctl reload nginx

```
(venv) swosti@newvm:/etc/nginx/sites-available$ sudo systemctl reload nginx
(venv) swosti@newvm:/etc/nginx/sites-available$ |
```

Or full restart:

sudo systemctl restart nginx

Check the domain you have configured to see apply changes on the domain you configured
http://swostiii.duckdns.org/



Lab-5: Virtualization

Objective: To install and configure virtualization software (VirtualBox) on a local machine and create multiple virtual machines.

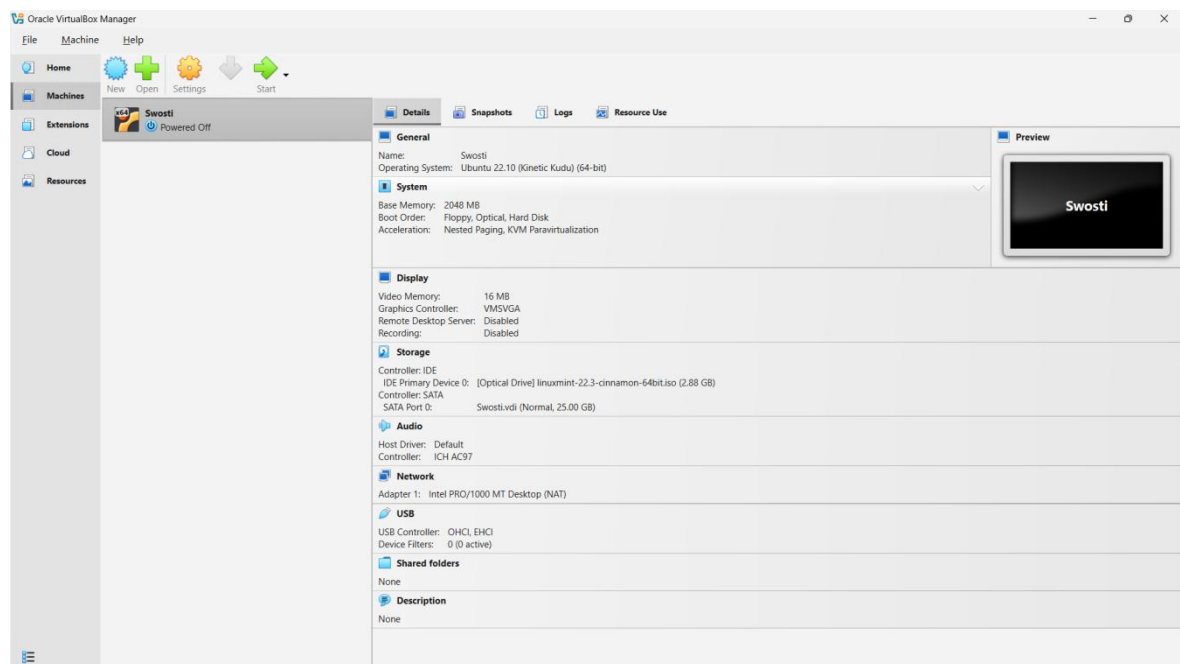
Tool Used:

➤ Oracle VirtualBox

1. Installing Linux Mint on VirtualBox

Step-1: Install VirtualBox

1. Download VirtualBox from the official website.
2. Install VirtualBox by following the on-screen instructions.
3. Open VirtualBox after installation.



Step-2: Create a New Virtual Machine

1. Click on “New” to create a new virtual machine.
2. A pop-up window will appear. Configure the following:
 - a) Name: Enter a name for the VM (e.g. “Linux Mint”).
 - b) Folder: Choose a location to store the VM files.
 - c) ISO Image: Select the Linux Mint ISO file.

New Virtual Machine

Virtual machine name and operating system

The ISO image is used to install the operating system on the VM.

VM Name: linux mint ✓

VM Folder: C:\Users\swosti\VirtualBox VMs

ISO Image: D:\Downloads\linuxmint-22.3-cinnamon-64bit.iso ✓✓

OS Edition: Linux

OS Distribution: Ubuntu

OS Version: Ubuntu 22.10 (Kinetic Kudu) (64-bit)

☒ Proceed with Unattended Installation

Detected OS type: Ubuntu 22.10 (Kinetic Kudu) (64-bit). VirtualBox will install the OS using an unattended installation when the VM is created. Supply the required information in the following steps.

Help Back Next Cancel

Step-3: Configure System Settings

1. Set up **username**, **password** and **hostname** for the VM.

New Virtual Machine

Set up unattended guest OS installation

Enter the information that will be required when the OS is installed.

User Name and Password:

User Name: swosti ✓

Password: •••••

Confirm Password: •••••

OS Installation Options:

Product Key: _____

Host Name: swosti ✓

Domain Name: myguest.virtualbox.org ✓

☐ Install in Background

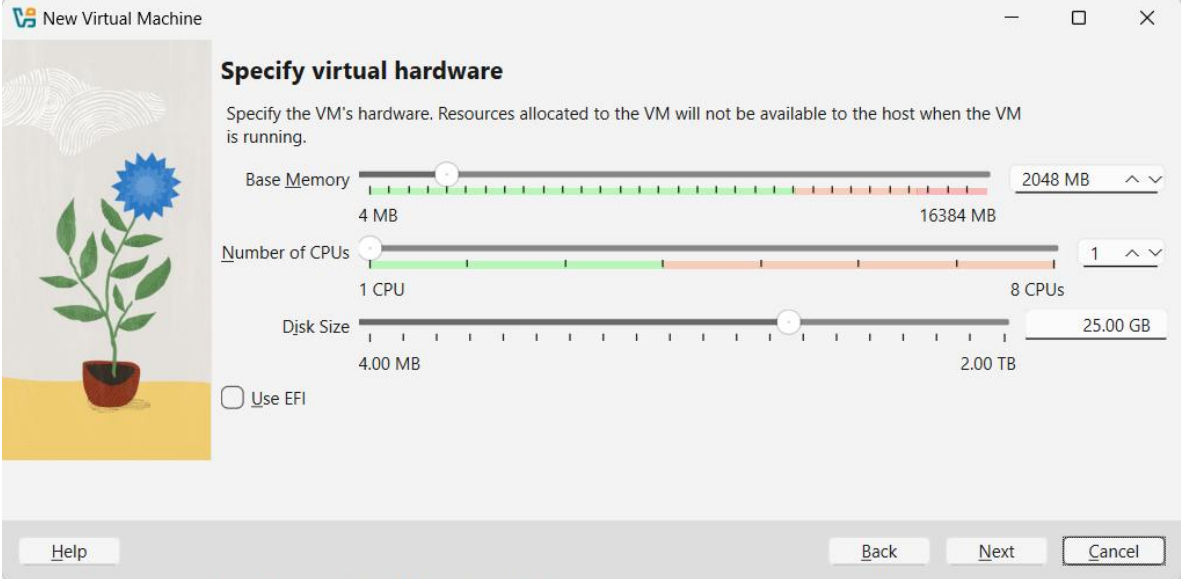
☐ Install Guest Additions

Guest Additions ISO Image: C:\Program Files\Oracle\VirtualBox\VBBoxGuestAdditions.iso

Help Back Next Cancel

Step-4: Customize hardware resources based on system requirements:

1. **Memory (RAM)** allocation
2. **Processor (CPU)** cores
3. **Storage capacity**



Specify virtual hardware

Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.

Base Memory: 2048 MB (Slider from 4 MB to 16384 MB)

Number of CPUs: 1 (Slider from 1 CPU to 8 CPUs)

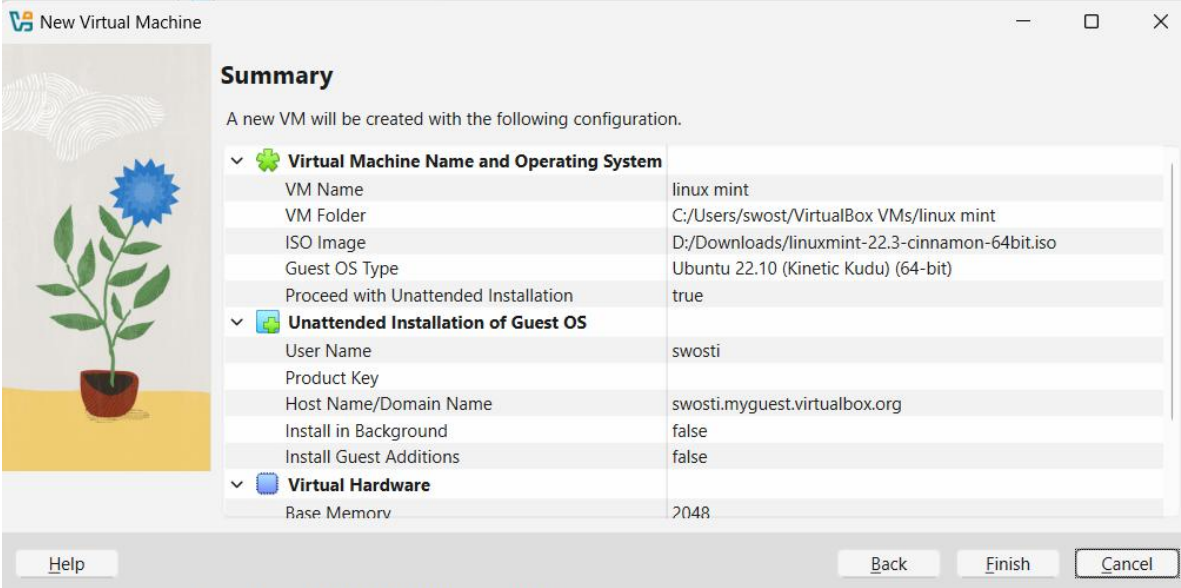
Disk Size: 25.00 GB (Slider from 4.00 MB to 2.00 TB)

☐ Use EFI

Buttons: Help, Back, Next, Cancel

Step-5:Finalizing Setup

1. Review the summary of the virtual machine configuration.
2. Click “**Finish**” to complete the setup.



Summary

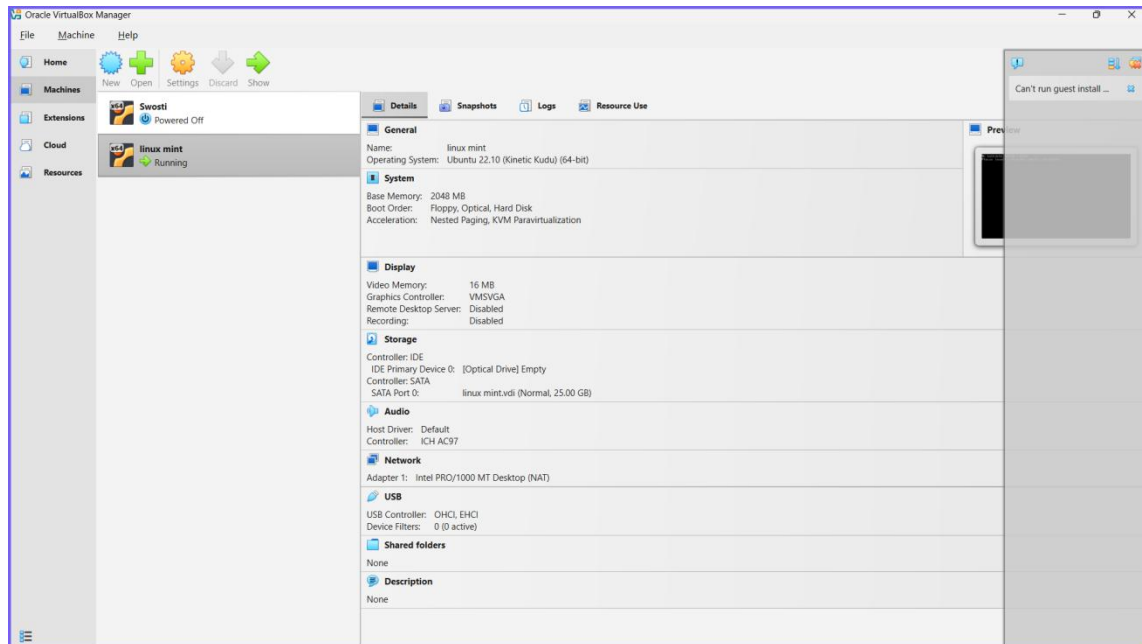
A new VM will be created with the following configuration.

- Virtual Machine Name and Operating System**
 - VM Name: linux mint
 - VM Folder: C:/Users/swosti/VirtualBox VMs/linux mint
 - ISO Image: D:/Downloads/linuxmint-22.3-cinnamon-64bit.iso
 - Guest OS Type: Ubuntu 22.10 (Kinetic Kudu) (64-bit)
 - Proceed with Unattended Installation: true
- Unattended Installation of Guest OS**
 - User Name: swosti
 - Product Key:
 - Host Name/Domain Name: swosti.myguest.virtualbox.org
 - Install in Background: false
 - Install Guest Additions: false
- Virtual Hardware**
 - Base Memory: 2048

Buttons: Help, Back, Finish, Cancel

Step-6: Installing Linux Mint

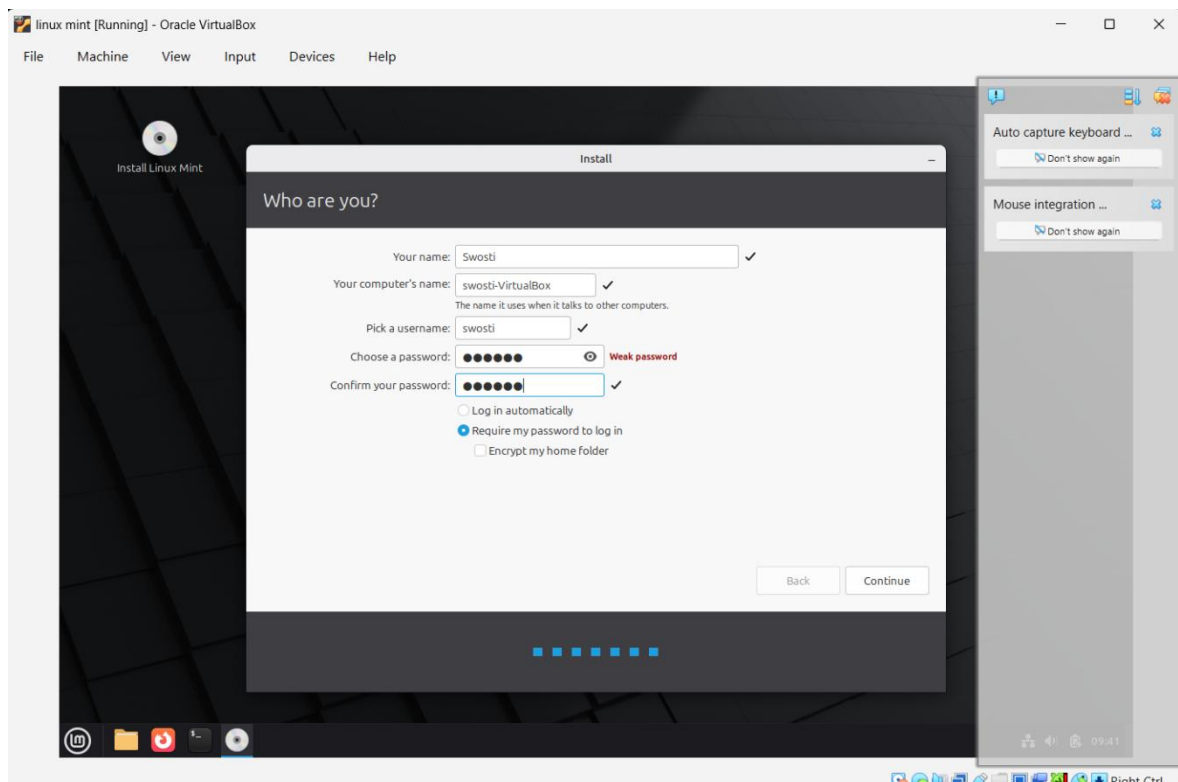
1. Start the virtual machine from the VirtualBox main interface.



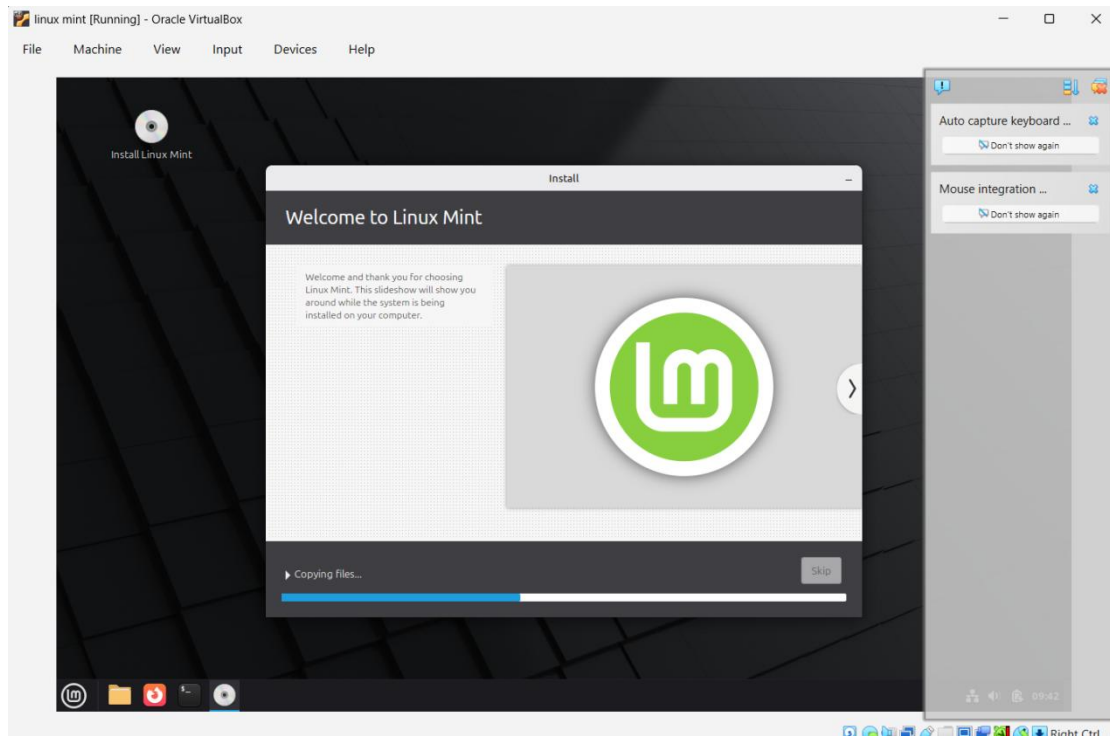
2. The installation process for Linux Mint will begin.

3. Follow on-screen instructions:

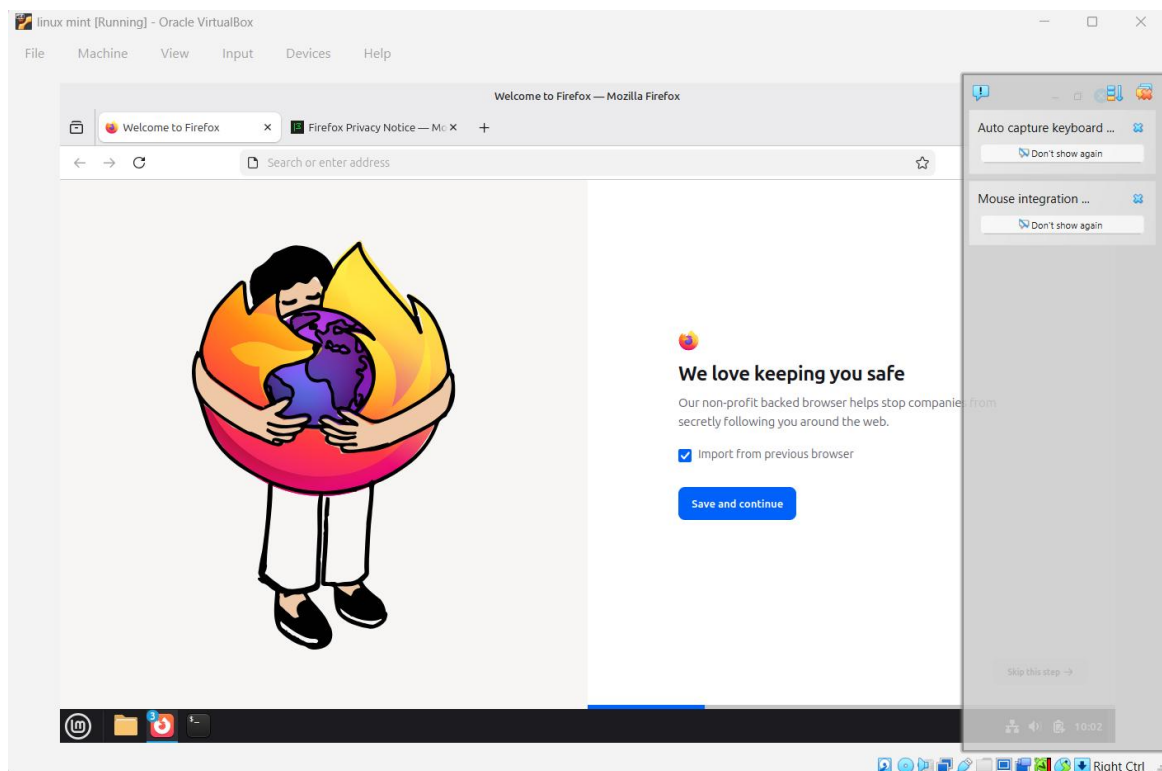
- a) Pick a **username** and **password**.
- b) Continue with the installation process.



4. Once completed, **Linux Mint** will successfully launch inside VirtualBox.



5. Successfully launch “Linux Mint” Operating system on Virtual Box.



Lab 6 MapReduce and Parallel Computing

Objective:

Implement and test the MapReduce programming model using Hadoop and Python, perform word count and analyze parallel efficiency and performance improvements.

Tools Used:

- JDK x64(Windows HotSpot)
- Hadoop 3.5.0
- Python 3.12+
- Windows Command Prompt(cmd.exe)

1. Check Java Installation

Installed OpenJDK 8, setup following the instruction and then checked the installed java version using Command Prompt.

a) Java Version

```
C:\Users\acer>java -version
openjdk version "1.8.0_502"
OpenJDK Runtime Environment (Temurin)(build 1.8.0_502-b07)
OpenJDK 64-Bit Server VM (Temurin)(build 25.502-b07, mixed mode)

C:\Users\acer>
```

b) Java Compiler Version

```
C:\Users\acer>javac -version
javac 1.8.0_502
```

2. Extract Hadoop

Download and then extract Apache Hadoop 3.5.0.

Create a new folder on desktop/download (e.g. MapReduce_Lab) and save these two files inside it using Notepad or VS Code:

mapper.py

```
import sys
```

```
for line in sys.stdin:
```

```
    line = line.strip().lower()
```

```
    words = line.split()
```

```
for word in words:  
    print(f'{word}\t1')
```

reducer.py

```
import sys  
current_word = None  
current_count = 0  
for line in sys.stdin:  
    line = line.strip()  
    if not line:  
        continue  
    try:  
        word, count = line.split('\t', 1)  
        count = int(count)  
    except ValueError:  
        continue  
    if current_word == word:  
        current_count += count  
    else:  
        if current_word:  
            print(f'{current_word}\t{current_count}')  
            current_word = word  
            current_count = count  
if current_word:  
    print(f'{current_word}\t{current_count}')
```

In the same folder, create a text file named input.txt and put some test text in it:

input.txt

```
cat  
dog  
cat  
cow  
parrot  
dog
```

Then, run these command

a) `cd <your folder location>`

b) `type input.txt | python mapper.py | sort | python reducer.py`

```
C:\Users\acer\Downloads\MapReduce_Lab>type input.txt | python mapper.py | so
rt | python reducer.py
cat      2
cow      1
dog      2
parrot   1
```

To evaluate execution performance across varying numbers of parallel processes(N) and dataset sizes (D), speedup and parallel efficiency are calculated as follows:

Speedup: $S_N = T_1 / T_N$

Where:

T_1 = execution time with 1 process/node

T_N = execution time with N process/node

Parallel Efficiency: $E_N = S_N / N * 100\%$

Where:

S_N = speedup achieved using N processes/ nodes

N = number of process/node